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A PYTHON PROGRAM TO
LOGISTIC REGRESSION USING SIGMOID FUNCTION

Aim:

To implement **Logistic Regression using the Sigmoid Function in Python** and visualize the probability curve for classification using the **Breast Cancer dataset**.

Requirements :

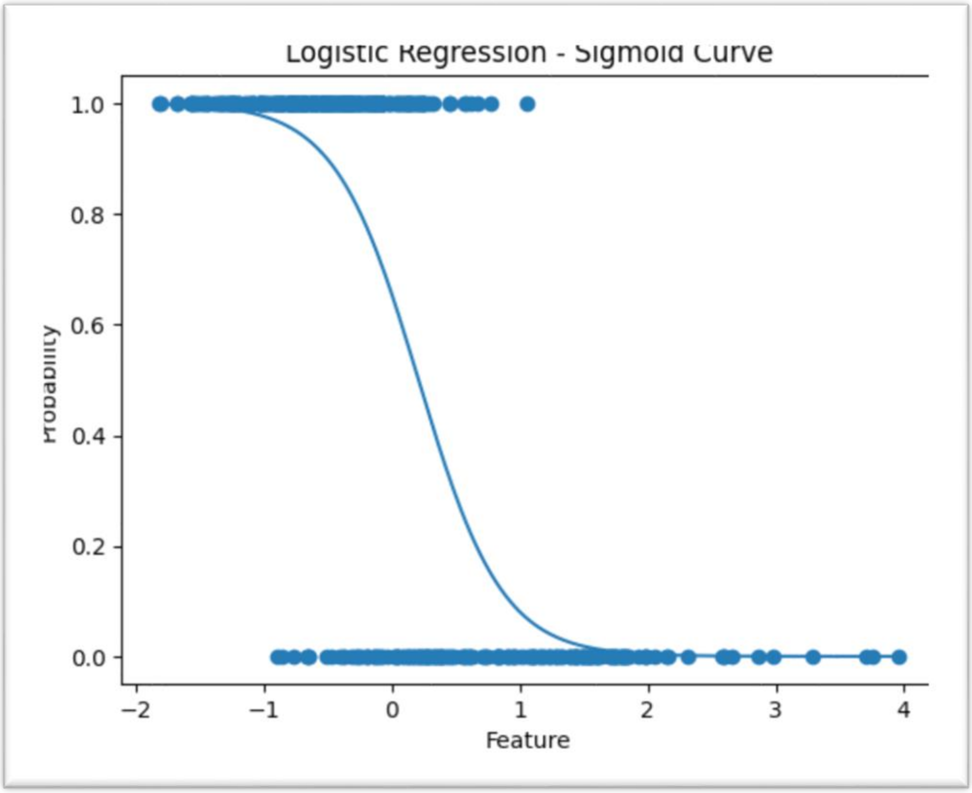
- Python 3.x
- Jupyter Notebook / Google Colab / Python IDE (VS Code, PyCharm, etc.)
- **NumPy** – for numerical operations
- **Matplotlib** – for data visualization
- **Scikit-learn (sklearn)** – for machine learning algorithms and dataset handling
- **Breast Cancer Dataset** available in sklearn.datasets

Program :

```
import numpy as np

import matplotlib.pyplot as plt

from sklearn.datasets import load_breast_cancer

from sklearn.linear_model import LogisticRegression

from sklearn.model_selection import train_test_split

from sklearn.preprocessing import StandardScaler

data = load_breast_cancer()
```

```
X = data.data[:, 0].reshape(-1, 1)
y = data.target
X_train, X_test, y_train, y_test = train_test_split(
X, y, test_size=0.2, random_state=42
)
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)
model = LogisticRegression()
model.fit(X_train, y_train)
x_values = np.linspace(X_train.min(), X_train.max(), 300).reshape(-1, 1)
y_prob = model.predict_proba(x_values)[:, 1]
plt.scatter(X_train, y_train)
plt.plot(x_values, y_prob)
plt.xlabel("Feature")
plt.ylabel("Probability")
plt.title("Logistic Regression - Sigmoid Curve")
plt.show()
```

Output :



Result:

In this program, **Logistic Regression** was implemented using the **Sigmoid function** to perform binary classification on the **Breast Cancer dataset**.

The dataset was first split into **training and testing sets**, and the feature values were **standardized using StandardScaler** to improve model performance. The **Logistic Regression model** was trained on the dataset, and the **sigmoid curve representing the probability of classification** was plotted using Matplotlib.

The sigmoid function maps the input values into a **probability range between 0 and 1**, making it suitable for **binary classification problems**. The visualization clearly shows how logistic regression predicts probabilities for different feature values.